

Pseudo-Orthogonal Group $O(s,t)$, $K = \mathbb{C}$ or $\mathbb{R}$

def: A symmetric Bilinear form: $Q: V \times V \rightarrow \mathbb{R}$

• Sym + $\mathbb{R}$ bilinear

• non-degenerate: $\forall v \neq 0 \in V \Rightarrow \exists w \in V$ with $Q(v, w) \neq 0$

ex: $\mathbb{R}^{s,t}$, w basis $e^1, \dots, e^{s+t}$ with standard symmetric bilinear form $\mathcal{Q}$

$$\mathcal{Q}(e_i, e_i) = +1, \quad \forall 1 \leq i \leq s$$

$$\mathcal{Q}(e_i, e_i) = -1, \quad \forall s+1 \leq i \leq s+t$$

$$\mathcal{Q}(e_i, e_j) = 0, \quad \forall i \neq j$$

$\rightarrow$ Signature of $\mathcal{Q}$ in $\mathbb{R}^{s,t}$ (s,t) $(\underbrace{+, \dots, +}_s, \underbrace{-1, \dots, -1}_t)$

$$\mathbb{R}^s = \mathbb{R}^{s,0}$$

$$\mathbb{R}^{1,s}, \mathbb{R}^{s,1} \cong \mathbb{M}^s$$

* An orthonormal Basis of $\mathbb{R}^{s,t}$: $v_1, \dots, v_s, w_1, \dots, w_t$

$$\mathcal{Q}(v_i, v_j) = \delta_{ij}, \quad \mathcal{Q}(v_i, w_j) = 0$$

$$\mathcal{Q}(w_i, w_j) = -\delta_{ij}$$

• $V \in \mathbb{R}^{s,t}$, $\|V\|_{\mathcal{Q}}^2 = \mathcal{Q}(V, V)$ ($\mathcal{Q}$ -norm squared)

Prop: Every non-degenerate symm bilinear form Q over $\mathbb{R}$ or $\mathbb{C}$ on V is isomorphic to precisely some $\mathbb{R}^{s,t}$ or $\mathbb{C}^d$

* Since we can multiply a vector in a $\mathbb{C}$ vector space by i and Q is complex bilinear

$$Q(v, v) = +1$$

$$\Rightarrow Q(iv, iv) = -1$$

Hence, there is no signature for SBF on a Complex vector space.

Def: $(V, \mathbb{R})$ w non deg symm form Q . The (pseudo)-orthog. Group

$$O(V, Q) = \{f \in GL(V) \mid Q(fv, fw) = Q(v, w) \quad \forall v, w \in V\}$$

* $\mathcal{Q}$ on $\mathbb{R}^{s,t}$ w/ (s,t)

$$O(s,t) \equiv \{A \in GL(s+t, \mathbb{R}) \mid \mathcal{Q}(Av, Aw) = \mathcal{Q}(v, w), \quad \forall v, w \in \mathbb{R}^{s+t}\}$$

Lorentz group: $O(1, t) \equiv O(s, 1)$

* Let $A \in \text{Mat}((s+t) \times (s+t), \mathbb{R})$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \quad \begin{array}{l} A_{11} \in \text{Mat}(s \times s, \mathbb{R}) \\ A_{12} \in \text{Mat}(s \times t, \mathbb{R}) \end{array} \quad \left| \quad \begin{array}{l} A_{21} \in \text{Mat}(t \times s, \mathbb{R}) \\ A_{22} \in \text{Mat}(t \times t, \mathbb{R}) \end{array} \right.$$

$$s \left\{ \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \right\}$$

* Then for η w signature (s, t)

$$\eta = \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix}$$

$$\eta(v, w) \mapsto v^T \eta w = (v_1 \dots v_s) \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \end{pmatrix}$$

and so, $\eta(Av, Aw) = \eta(v, w)$

$$(Av)^T \eta Aw = v^T \eta w = v^T A^T \eta Aw$$

$$\text{or } v^T [A^T \eta A - \eta] w = 0$$

$$\therefore A \in O(s, t) \text{ iff } A^T \eta A = \eta$$

$$A^T \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix} A = \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix}$$

$$\bullet \det(A^T \eta A) = \det(\eta)$$

$$| \det(A) |^2 = 1 \Rightarrow \det(A) = \pm 1$$

* The matrices $A \in O(s, t)$ satisfy $\nearrow$, If $A \in \text{Mat}((s+t) \times (s+t), \mathbb{R})$

$$A = (v_1, \dots, v_s, w_1, \dots, w_t), \quad v_i, w_j \in \mathbb{R}^{s+t}, \text{ then } A \in O(s, t) \text{ iff:}$$

$$\eta(v_i, v_j) = \delta_{ij}$$

$$\eta(w_i, w_j) = -\delta_{ij}$$

$$\eta(v_i, w_j) = 0$$

i.e. the vectors $v_1, \dots, v_s, w_1, \dots, w_t$ form an orthonormal basis for $\mathbb{R}^{s+t}$

proof: ($\Rightarrow$) Suppose $A = (v_1, \dots, v_s, w_1, \dots, w_t)$ and $A \in O(s, t)$

$$\bullet \text{ Then we know } A^T \eta A = \eta$$

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^T \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix}$$

$$= \begin{pmatrix} A_{11}^T & A_{21}^T \\ A_{12}^T & A_{22}^T \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ -A_{21} & -A_{22} \end{pmatrix} = \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix}$$

$$= \begin{pmatrix} A_{11}^T A_{11} - A_{21}^T A_{21} & A_{11}^T A_{12} - A_{21}^T A_{22} \\ A_{12}^T A_{11} - A_{22}^T A_{21} & A_{12}^T A_{12} - A_{22}^T A_{22} \end{pmatrix}$$

$1+3 \times \quad +3)$ $s=1, t=3$

$$\star \mathcal{O}(v_i, v_j) = \delta_{ij}$$

$$v_i^T \begin{pmatrix} I_s & 0 \\ 0 & -I_t \end{pmatrix} v_i = \begin{pmatrix} I_s & 0 \\ 0 & I_t \end{pmatrix}$$

$$\sum_{i=1}^s (v_i)^2 - \sum_{l=1}^t (v_l)^2 = 1$$

• Which is exactly what we have above.

Hence, if $A \in O(s, t)$ $A = (v_1, \dots, v_s, w_1, \dots, w_t)$

$$\begin{aligned} \text{then } \mathcal{O}(v_i, v_j) &= \delta_{ij} \\ \mathcal{O}(w_i, w_j) &= -\delta_{ij} \end{aligned} \quad \left| \quad \mathcal{O}(v_i, w_l) = 0 \right.$$

($\Leftarrow$) Suppose $v_1, \dots, v_s, w_1, \dots, w_t$ form an orthonormal basis of $\mathbb{R}^{s+t}$, then

$$\mathcal{O}(v_i, v_j) = \delta_{ij}$$

$$v_i^T \mathcal{O} v_j = \delta_{ij}$$

$$w_i^T \mathcal{O} w_j^T = -\delta_{ij} \quad \square$$

Lemma: $\exists$ a canonical Iso morphism $O(s, t) \cong O(t, s)$

Proposition: (i) $O(s, t)$ is a Linear Lie group

(ii) If $s, t \neq 0$, then $O(s, t)$ is not compact

(iii) $\Delta(s, t)$, then as a complex L.A. $\Delta(s, t) \otimes_{\mathbb{R}} \mathbb{C} \cong \Delta(s+t) \otimes_{\mathbb{R}} \mathbb{C}$

$$\dim(O(s, t)) = \dim(O(s+t)) = \frac{1}{2}(s+t)(s+t-1)$$

$$\star n \equiv s+t, \quad n^2 - \frac{1}{2}(n^2 - n) + n$$

$$= n^2 - \frac{1}{2}[n^2 - n + 2n] = n^2 - \frac{1}{2}[n^2 + n] = \frac{n}{2}[2n - n - 1]$$

$$\therefore \dim = \frac{n}{2}(n-1) = \frac{(s+t)}{2}(s+t-1) \quad \checkmark$$

* $O(0, n) \cong O(n, 0) = O(n)$:

Has two connected components

$$O(n) = \underbrace{(\pm 1 \det)}_U U(-1 \det)$$

$$SO(n)$$

* # of connected components of $O(s, t)$:

$$V_+ = \text{span}\{e_1, \dots, e_s\}$$

$$V = V_+ \oplus V_-$$

$$V_- = \text{span}\{e_{s+1}, \dots, e_{s+t}\}$$

$$\pi: V \rightarrow V_+$$

• $W \subset V$ be a maximally η -positive def. vec. subspace of dim s .

$$\text{Then } \pi|_W: W \rightarrow V_+$$

is an isomorphism.

$$\text{Kern}(\pi) = \{w \in W \mid \pi(w) = 0\}$$

Note if $w \neq 0$, then, $\eta(w, w) > 0$

$$\eta(\pi(w), \pi(w)) = +1$$

But Kern is trivial.

Def: Fix an orientation on V_+ , then $\exists$ V maximally positive def. vec. sub.

$W \subset V$, a unique orientation s.t. $\pi|_W$ is orientation preserving

Supp. $A \in O(s, t)$, since A preserves η , it maps pos. def. subsp. of V to (maximally) positive def. subspaces

Def: time-orientability: of $A \in O(s, t)$ to be $+1$ or -1 depending on whether:

$$A|_{V_+}: V_+ \rightarrow A(V_+)$$

preserves or does not orientation.

Lemma: Let $A \in O(s, t)$, $W \subset V$ maximally positive definite subspace. Then

A has time orientability $+1$ iff $A|_W: W \rightarrow A(W)$

preserves orientation $\circ \circ \circ$

with orientation on W and $A(W)$ chosen

A map preserves orientation means

$$\det(A) > 0$$

Use the projection π .

Proof: ($\Rightarrow$) Suppose $A \in O(s, t)$, $W \subset V$ chosen.

$\pi|_W: W \rightarrow V_+$ is an isomorphism, we $\forall$ a basis $w_1, \dots, w_s$ of W

$$w_i = e_i + \bar{v}_i \quad | \quad \bar{v}_i \in V_-, \quad e_i \in V_+$$

• Define $\alpha = \sum_{i=1}^s a_i \bar{v}_i$ $(a_1, \dots, a_s) \in \mathbb{R}^s \setminus \{0\}$

by construction $q(\alpha, \alpha) < 0$
 $\Rightarrow W = \sum_{i=1}^s q_i w_i = a^i e_i + a^i w_i$

$q(w, w) > 0$ (by design)

$q(w, w) = a^i q_i + q(\alpha, \alpha) > 0 \Rightarrow \sum_{i=1}^s q_i^2 + q(\alpha, \alpha) > 0$

• For $\tau \in [0, 1]$, $W_\tau = \text{span} \{e_1 + \tau \bar{v}_1, \dots, e_s + \tau \bar{v}_s\}$

as above, W_τ is maximally positive def. $\forall \tau \in [0, 1]$

$W_0 = V_+$

$W_1 = W$

• Let $A_\tau = A|_{W_\tau}$

$W_\tau \xrightarrow{A_\tau} A(W_\tau)$

$\downarrow \pi$

$V_+ \xrightarrow{B_\tau} V_+$

$w \in W_\tau \Rightarrow w = \sum_{i=1}^s a_i (e_i + \tau \bar{v}_i)$

$\pi(w) = \sum_{i=1}^s a_i e_i$

$A|_{W_\tau} := \{a_i (Ae_i + \tau A\bar{v}_i) \mid \dots\}$

$\pi \circ A_\tau = B_\tau \circ \pi$

* $\det(A) > 0$ iff $\det(B_0) > 0$

• Since the determinant is continuous and B_τ is a continuous curve in $GL(V_+)$

B_0 has positive det. iff B_1 has positive det.

Def: $SO(s, t) = \{A \in O(s, t) \mid \det(A) = 1\}$ (a)

$O^+ = \{A \in O(s, t) \mid A \text{ has time orientability } +1\}$ (b)

$SO^+(s, t) = SO(s, t) \cap O^+(s, t)$ (c)

(a) ..

(b) Orthochronous pseudo-orth group

(c) proper-orthochronous

* $A \in O(s, t) \cong \text{Mat}(\mathbb{R}(t) \times (s+t), \mathbb{R}) \mid \dots$

$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

$O^+(s, t) = \{A \in O(s, t) \mid \det(A_{11}) > 0\}$

$SO^+(s, t) = \{A \in O(s, t) \mid \det(A) = 1 \text{ \& } \det(A_{11}) > 0\}$

* For $O(n)$: time orientation = determinant

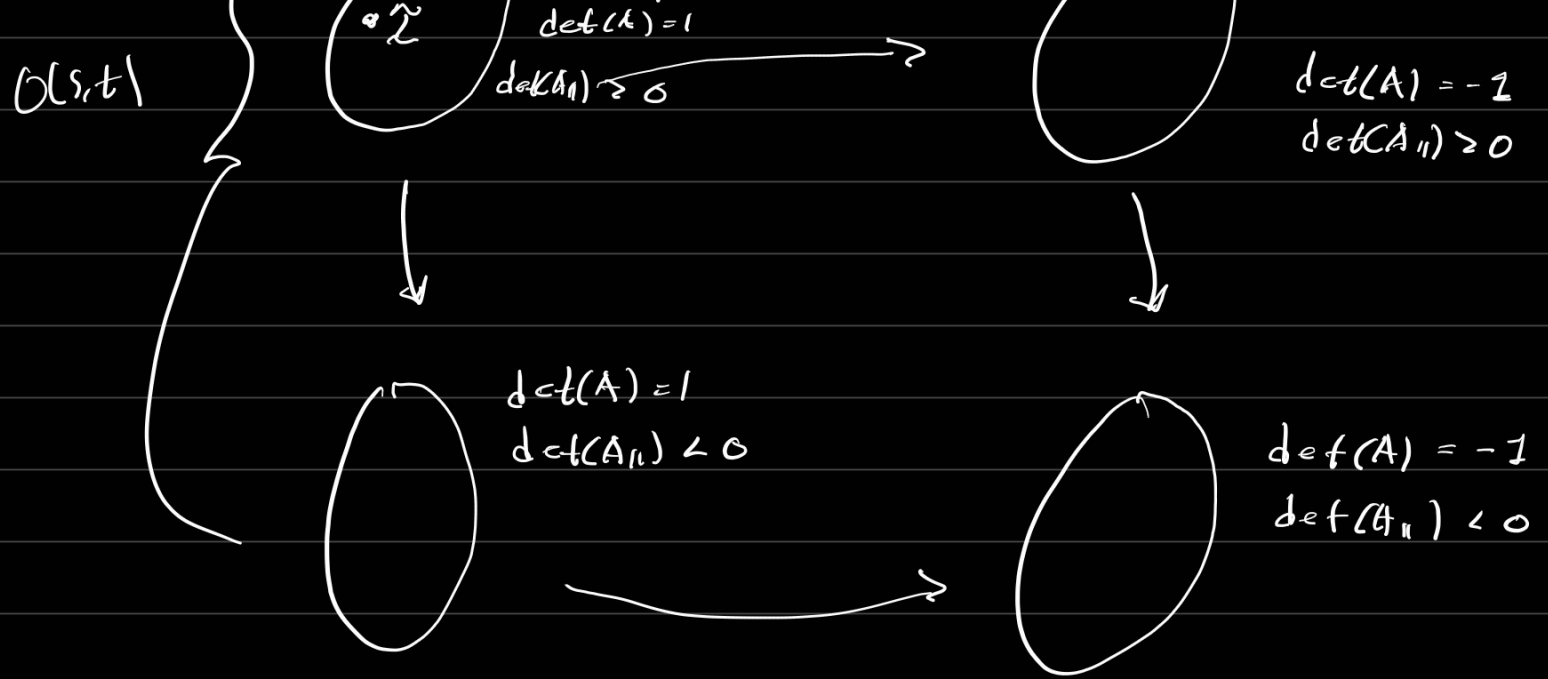
$O^+(n) = SO(n) = SO^+(n)$

Proposition: [connected Comp. of the Identity of Pseudo-orthogonal Groups]

$SO^+(s, t)$ is the connected component of the identity in $O(s, t)$

• If $s, t \neq 0$ then $\exists$ 4 connected components

$SO^+(s, t)$



* From the isomorphism $O(s, t) \cong O(t, s)$, time orientability is also given by $SO^+(s, t) = \{A \in O(s, t) \mid \det(A) = 1 \text{ \& \det}(A_{22}) > 0\}$

* As a smooth manifold: $(s, t \geq 1)$

$$SO^+(s, t) \cong SO(s) \times SO(t) \times \mathbb{R}^{st}$$

as $SO(s) \times SO(t)$ is the maximal compact subgroup of $SO^+(s, t)$ which determines the fundamental group of $SO^+(s, t)$

* $SO^+(1, 3) \cong SO(3) \times \mathbb{R}^3$

$\hookrightarrow$ Has a fundamental group $\mathbb{Z}_2$ | $SL(2, \mathbb{C})/\mathbb{Z}_2 = SO(3)$

Clifford Algebras:

Def: An associative $\mathbb{K}$ -alg. with unit 1 is a $\mathbb{K}$ vec. space A of finite dim together w a bilinear, associative product.

$$\cdot : A \times A \rightarrow A$$

$$1 \in A \quad | \quad 1 \cdot a = a = a \cdot 1 \quad \forall a \in A$$

$\cdot$: distributive & associative

The direct sum: of assoc. algebras A, B : $A \oplus B$

$$(a, b) \cdot (a', b') = (a \cdot_A a', b \cdot_B b') \quad \forall a, a' \in A, b, b' \in B$$

Tensor product: $A \otimes_{\mathbb{K}} B$

$$(a \otimes b) \cdot (a' \otimes b') = (a \cdot_A a') \otimes (b \cdot_B b')$$

Isomorphism: bijective homomorphism

Def: Let V be a $\mathbb{K}$ vec. space w Q (V, Q) . A Clifford algebra of (V, Q) is a pair $(Cl(V, Q), \gamma)$ where:

1. $Cl(V, Q)$ is an associative K algebra w unit 1

2. Clifford Relation: $\gamma: V \rightarrow Cl(V, Q)$ is a linear map

$$\{\gamma(v), \gamma(w)\} = -2Q(v, w) \cdot 1 \quad \forall v, w \in V$$

3. Universal property: If A is some other K -alg. w unit 1 and

$$\delta: V \rightarrow A \text{ a linear map w } \{\delta(v), \delta(w)\} = -2Q(v, w) \cdot 1$$

then $\exists!$ algebra homo $\phi: Cl(V, Q) \rightarrow A$ s.t.

$$V \xrightarrow{\gamma} Cl(V, Q)$$

$$\searrow \delta \quad \downarrow \phi$$

$$A$$

$$\delta = \phi \circ \gamma$$

* Can think of γ as "linear square roots" of $-Q$

$$\gamma(v)^2 = -Q(v, v) \cdot 1 \quad \forall v \in V$$

$$* D^2 = D \circ D = \Delta = -Q_{ij} \partial_i \partial_j$$

$$\text{formally, if we let } D = \gamma(e_i) \frac{\partial}{\partial x_i}$$

$$D \circ D = (\gamma(e_i) \partial_i)(\gamma(e_j) \partial_j)$$

$$= \frac{1}{2} (\gamma(e_i) \gamma(e_j) + \gamma(e_j) \gamma(e_i)) \partial_i \partial_j$$

$$= -Q_{ij} \partial_i \partial_j$$

$$\therefore \gamma_i \gamma_j + \gamma_j \gamma_i = -2Q_{ij} \cdot 1 \quad (\text{as above})$$

$$\text{Rep of } Cl(V, Q): \quad \rho: Cl(V, Q) \rightarrow \text{End}(\Delta) \text{ over } \mathbb{C}$$

* If $v \in V$ w $Q(v, v) = \pm 1$ then $\forall x \in V$

$$\pm \gamma(v) \cdot \gamma(x) \cdot \gamma(v) = \begin{cases} -\gamma(x), & \text{if } x \parallel v \\ +\gamma(x), & \text{if } x \perp v \end{cases}$$

$$\text{proof: } \gamma_v \gamma_x = -2Q_{vx} - \gamma_x \gamma_v$$

$$= \pm (-2Q_{vx} - \gamma_x \gamma_v) \gamma_v$$

$$= \pm (-2Q_{vx} \gamma_v - \gamma_x (\pm 1))$$

$$= \mp (2Q_{vx} \gamma_v \pm \gamma_x)$$

$$\parallel: \mp (2\gamma_x \pm \gamma_x) \dots$$

* γ is isometric, $\forall v \in V$ s.t. $Q(v, v) = \pm 1$ act as reflections in the hyperplane $v^\perp$: arbitrary products of such vectors act as pseudo-orth. transformations.

Existence & Uniqueness of $Cl(V, Q)$:

Thm: Existence of Clifford Algebras: $\forall$ finite dim $\mathbb{K}$ -v.s. $(V, Q) \exists$ $(CL(V, Q), \gamma)$.

Proof: $T(V)$ tensor algebra of V

$$T(V) = \bigoplus V^{\otimes n} = \mathbb{K} \oplus V \oplus (V \otimes V) \oplus \dots$$

$I(Q)$ denote the $^{n \geq 0}$ two sided ideal in $T(V)$ generated by

$$\{v \otimes v + Q(v, v) \cdot 1 \mid v \in V\}$$

$$\bullet \quad I, V \Rightarrow a \cdot v \in I$$

Then $\exists \sim$ s.t. $v, w \in T(V)$

$$v - w \in I(Q)$$

Define: $CL(V, Q) = T(V) / I(Q)$

CL is then an associative algebra with product.

$$a, b \in T(V)$$

$$[a], [b] \in CL(V, Q)$$

$$[a] \cdot [b] = [a \otimes b]$$

$\bullet$ If $Q = 0$, then $CL(V, 0) = \wedge^* V$ (exterior Algebra)

$$\bullet \quad i: V \rightarrow T(V)$$

$$\pi: T(V) \rightarrow CL(V, Q)$$

$$\gamma = \pi \circ i$$

$$I(Q) \cong \{0\}$$

is linear:

$$\begin{aligned} \gamma(av + bw) &= \pi \circ i(av + bw) \quad (\text{Hom}) \\ &= a[v] + b[w] \end{aligned}$$

$$\gamma(v) \cdot \gamma(v) = [v] \cdot [v] = [v \otimes v] = -Q(v, v) 1$$

and so $\{\gamma(v), \gamma(w)\} = -2Q(v, w) 1$, $\gamma(V)$ generates $CL(V, Q)$ mult.

By def: every linear map $\delta: V \rightarrow A$ to an associative algebra A extends to an algebra homo

$$\Delta: T(V) \rightarrow A$$

If $\{\delta(v), \delta(w)\} = -2Q(v, w) 1$, then $I(Q)$ is a subset of $\ker(\Delta)$ and Δ descends to a homo

$$\phi: CL(V, Q) \rightarrow A$$

$$\phi \circ \gamma = \delta$$

$$\gamma = \pi \circ i$$

$$V \xrightarrow{\gamma} CL$$

$$\searrow \delta \quad \downarrow \phi$$

$$\leadsto \phi \circ \pi(v_1 \otimes \dots \otimes v_k)$$

$$= \phi(\pi(i(v_1) \dots i(v_k)))$$

$$= \phi(Y(v_1)) \dots \phi(Y(v_k))$$

$$= \delta(v_1) \dots \delta(v_k)$$

$$\Rightarrow \Delta(v_1 \otimes v_2 \otimes \dots \otimes v_k) = \delta(v_1) \dots \delta(v_k)$$

$$v_i, w \in V$$

$$\Delta(v \otimes w + w \otimes v + 2Q(v, w)1)$$

$$= \Delta(i(v) \otimes i(w) + i(w) \otimes i(v) + 2Q(v, w)1)$$

$$= \Delta(i(v)) \Delta(i(w)) + \Delta(i(w)) \Delta(i(v)) + 2Q(v, w)1$$

$$= \delta(v) \delta(w) + \delta(w) \delta(v) + 2Q(v, w)1$$

$$= 0$$

Thus Δ descends to ϕ

Uniqueness: $(CL(V, Q), \gamma) \cong (CL(V, Q), \gamma') \exists f$ $A = CL'$
 $f: CL \rightarrow CL'$ s.t. $f \circ \gamma = \gamma'$ $f = \phi$

* Image of V under γ generates CL multiplicatively

* If $Q \equiv 0$, then $\exists$ an algebra isomorphism

$$(CL(V, 0), \cdot) \cong (\wedge^\bullet V, \wedge)$$

where γ is given by the standard embedding $V \hookrightarrow \wedge^1 V$

proof: $Q=0$, thus

$$\gamma(v) \cdot \gamma(w) + \gamma(w) \cdot \gamma(v) = 0 \quad \forall v, w \in V$$

$$\bullet = \wedge : \quad \gamma(v) \wedge \gamma(w) = -\gamma(w) \wedge \gamma(v)$$

$$v \in V \Rightarrow v = v^i e_i \mapsto v_i dx^i$$

$$n = \dim(V), \quad A = \wedge^\bullet V = \bigoplus_{k=0}^n \wedge^k V$$

Corollary: Suppose $\dim_{\mathbb{K}} V = n$ and $e_1, \dots, e_n$ is an ortho. norm basis

for (V, Q) , Then $\{\gamma(e_{i_1}) \cdot \gamma(e_{i_2}) \dots \gamma(e_{i_k})\}$

$$1 \leq i_1 < i_2 < \dots < i_k \leq n, \quad 0 \leq k \leq n, \quad k=0 \Rightarrow 1$$

Span $CL(V, Q)$ as a vector space. This implies that

$$\dim(CL(V, Q)) \leq 2^n$$

$$\gamma_i \equiv \gamma(e_i)$$

proof: $\{e_{i_1} \otimes \dots \otimes e_{i_k} \}_{1 \leq i_1 < \dots < i_k \leq n, k \in \mathbb{N}_0}$ is a basis for $T(V)$

and so $\{\gamma_{i_1} \cdot \dots \cdot \gamma_{i_k}\}_{1 \leq i_1 < \dots < i_k \leq n, k \in \mathbb{N}_0}$ generates $T(V)/I$ as a vec.

space. Can use the polarity relation $\{\gamma_i, \gamma_j\} = -2Q(e_i, e_j)1$

to order the set on $T(V)/I$

$$\{\gamma_{i_1} \cdot \gamma_{i_2} \cdot \dots \cdot \gamma_{i_k}\}_{1 \leq i_1 < i_2 < \dots < i_k \leq n, k \in \mathbb{N}_0}$$

$$\begin{cases} \dim(\Lambda^0 V) = 2^n \\ \dim(\Lambda^2 V) = \frac{1}{2}(n^2 - n) \quad , \quad \dim(\Lambda^k V) = \binom{n}{k} \\ \dim(\Lambda^* V) = \sum_{k=0}^n \binom{n}{k} = 2^n \end{cases}$$

examples: $V = \mathbb{R}$, $\mathcal{Q}(x, y) := x \cdot y$, $\text{span } e_1 = \mathbb{R}$

• 1 and e_1 generate $CL(\mathbb{R}, \mathcal{Q}) = \text{span} \{1, e_1\}$

$\leadsto$ Assume $e_1 = \alpha 1$ for $\alpha \in \mathbb{R}$, then

$$\alpha^2 \cdot 1 = e_1^2 = -\mathcal{Q}(e_1, e_1) \cdot 1 = -1$$

a contradiction.

• The vect. space is $\phi: CL(\mathbb{R}, \mathcal{Q}) \rightarrow \mathbb{C}$ defined by

$$\phi(\alpha 1 + \beta e_1) := \alpha \phi(1) + \beta \phi(e_1) = \alpha \cdot 1 + \beta i$$

and is also an algebra Homo. $CL(\mathbb{R}, \mathcal{Q}) \cong \mathbb{C}$

* Consider $(\mathbb{R}, \tilde{\mathcal{Q}})$ with $\tilde{\mathcal{Q}}(x, y) = -x \cdot y$, then

$$e_1^2 = -\tilde{\mathcal{Q}}(e_1, e_1) \cdot 1 = 1$$

• $1, e_1 \in CL(\mathbb{R}, \tilde{\mathcal{Q}})$, $\exists \alpha \in \mathbb{R}$ $e_1 = \alpha \cdot 1$, then

$$e_1 - \alpha \cdot 1 = x \otimes (e_1 \otimes \alpha 1 - 1) \otimes y \quad , \quad \text{for some } x, y \in T(\mathbb{R})$$

meaning, as an element of $T(\mathbb{R})$ $e_1 - \alpha 1 \in \mathcal{I}$

where $\tilde{\mathcal{Q}}(e_1, e_1) = -\alpha$

$\mathbb{Z}$ -graded Tensor Algebra: $T(V) = \bigoplus_{i \in \mathbb{Z}} T^i(V)$

$$T^i(V) \cdot T^j(V) \subset T^{i+j}(V)$$

$$T^i(V) = \begin{cases} V^{\otimes i} & i \geq 0 \\ 0 & i < 0 \end{cases}$$

$\mathbb{Z}_2$ -grading: $T(V) = T^0(V) \oplus T^1(V)$

$$T^0(V) \equiv \bigoplus_{i \text{ even}} T^i(V)$$

$$T^1(V) \equiv \bigoplus_{i \text{ odd}} T^i(V)$$

which descends to the Clifford Algebra $CL(V, \mathcal{Q})$ since $\mathcal{I}$ is generated by elements in the even part.

• Let $\gamma: V \rightarrow CL(V, \mathcal{Q})$ and consider $\tilde{\gamma}: V \rightarrow CL(V, \mathcal{Q})$

$$\tilde{\gamma}(v) := -\gamma(v) \quad , \quad \forall v \in V$$

$$\tilde{\gamma}(v)^2 = \gamma(v)^2 = -B(v, v) 1$$

Thus, $\exists$ an algebra homo $\phi: CL \rightarrow CL$ $\phi \circ \gamma = \tilde{\gamma}$

$$\phi 1_0 = -\tilde{\gamma}$$

$$\gamma \rightarrow CL \downarrow \phi$$

which yields a decomposition $V \xrightarrow{\phi} \mathbb{C}l$

$$\mathbb{C}l(V, Q) = \mathbb{C}l^0(V, Q) \oplus \mathbb{C}l^1(V, Q)$$

$\mathbb{C}l^0(V, Q)$: +1 - eigenspace of ϕ

$\mathbb{C}l^1(V, Q)$: -1 - eigenspace of ϕ

Let $m \in \mathbb{C}l(V, Q)$, then $\phi m = m \Rightarrow m \in \mathbb{C}l^0(V, Q)$
 $\phi m = -m \Rightarrow m \in \mathbb{C}l^1(V, Q)$

Def: $\mathbb{C}l^0(V, Q) = T^0(V) / (T^0(V) \cap I(Q))$

$\mathbb{C}l^1(V, Q) = T^1(V) / (T^1(V) \cap I(Q))$

$I(Q) = (T^0(V) \cap I(Q)) \oplus (T^1(V) \cap I(Q))$

$\mathbb{C}l^i \cdot \mathbb{C}l^j \subset \mathbb{C}l^{i+j \pmod{2}}$

$\mathbb{C}l^0(V, Q)$ is a subalgebra of $\mathbb{C}l$ and is spanned by products $v_1 \dots v_m$ of an even # of vectors $v_i \in V$

Cliff + Exterior Alg:

Def: $v \in V$, $\sigma \in \wedge^k V$, $\exists$ a unique $(k-1)$ -form $v \lrcorner \sigma \in \wedge^{k-1} V$ called a contraction of v and σ s.t.

(i) If $\sigma \in V$, $v \lrcorner \sigma = Q(v, \sigma)$

(ii) $\forall \sigma \in \wedge^k V$, $\omega \in \wedge^l V$

$v \lrcorner (\sigma \wedge \omega) = (v \lrcorner \sigma) \wedge \omega + (-1)^k \sigma \wedge (v \lrcorner \omega)$

Thm: $\mathbb{C}l(V, Q) \cong_{\text{vec}} \wedge^* V$

$\exists$ a canonical iso of vec. spaces $\wedge^* V \rightarrow \mathbb{C}l(V, Q)$

In any orthonormal basis $e_1, \dots, e_n$ of (V, Q) , this is given by $e_{i_1} \wedge e_{i_2} \wedge \dots \wedge e_{i_k} \mapsto \gamma(e_{i_1}) \cdot \gamma(e_{i_2}) \cdot \dots \cdot \gamma(e_{i_k})$

$\dim(\mathbb{C}l(V, Q)) = 2^n$

Proof: $\delta: V \rightarrow \text{End}(\wedge^* V)$

$\delta(v): \wedge^* V \rightarrow \wedge^* V$

$\alpha \mapsto v \wedge \alpha - v \lrcorner \alpha$

$\{\delta(v), \delta(w)\} \alpha = \dots$

$\delta(v) \delta(w) \alpha = \delta(v) [w \wedge \alpha - w \lrcorner \alpha]$

$= v \wedge w \wedge \alpha - v \lrcorner (w \wedge \alpha) - v \wedge (w \lrcorner \alpha) + v \lrcorner (w \lrcorner \alpha)$

$= v \wedge [w \wedge \alpha - w \lrcorner \alpha] - v \lrcorner [w \wedge \alpha - w \lrcorner \alpha]$

$\dots = -2 Q(v, w) \alpha$

Thus, $\delta(v) \in \text{End}(\wedge^* V)$ satisfy the Clifford conditions.

Extends to an algebra homo

$$\phi: CL(V, \Omega) \rightarrow \text{End}(\wedge^\bullet V)$$

$$\phi(\gamma(v)) \mapsto \delta(v)$$

$$f: CL \rightarrow \wedge^\bullet V$$

$$x \mapsto \phi(x)(1_{\wedge^\bullet V})$$

$$f(\gamma(e_{i_1}) \cdots \gamma(e_{i_n})) = e_{i_1} \wedge \cdots \wedge e_{i_n}$$

f : symbol map

f^{-1} : Quantization map

$$\begin{array}{ccc} CL & \xrightarrow{\phi} & \text{End}(\wedge^\bullet V) \\ & \searrow f & \downarrow \text{id} \\ & & \wedge^\bullet V \end{array}$$

* Linear map γ is an Embedding: $\gamma: V \rightarrow CL(V, \Omega)$
 $\leadsto$ Identify V with its image under γ .

$$(s, t): \mathbb{R}^{s+t}, CL(s, t) \quad (V, \Omega) = (\mathbb{R}^{s+t}, \Omega)$$

$$(s, t) = (n, 0) : CL(n)$$

$$(V, \Omega) = (\mathbb{C}^d, q) : CL(q)$$

Complexified CL Alg: $\exists$ an iso of complex associative algs.

$$\mathbb{C} CL(s, t) \cong CL(s, t) \otimes_{\mathbb{R}} \mathbb{C}$$

Complex reps of CL are equiv. to "s" of $CL(s, t)$

Proof: $\delta: \mathbb{C}^{s+t} \cong \mathbb{R}^{s+t} \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow CL(s, t) \otimes_{\mathbb{R}} \mathbb{C}$

$$(v \otimes z) \mapsto \gamma(v) \otimes z$$

Note that $\delta(v \otimes z) = \gamma(v) \otimes z$

$$\begin{aligned} \leadsto \delta(v \otimes z) \cdot \delta(v \otimes z) &= (\gamma(v) \otimes z) \cdot (\gamma(v) \otimes z) = \gamma(v)^2 \otimes z^2 \\ &= -q(v, v) 1 \otimes z^2 \\ &= -q(v \otimes z, v \otimes z) \cdot 1 \end{aligned}$$

Even $\mathbb{C}$ -CL: For $\forall n \geq 1$

$$\mathbb{C} L^0(n) \cong \mathbb{C} L(n-1)$$

$$e_n = (0, \dots, \underset{n}{1}, \dots)$$

Proof: e_n : n^{th} basis vec. of $\mathbb{C}^n$

$$\delta: \mathbb{C}^{n-1} \longrightarrow \mathbb{C} L^0(n)$$

$$x \mapsto x \cdot e_n$$

$$\delta x^2 = x \cdot e_n \cdot x \cdot e_n$$

$$= x \cdot x$$

$$= -q(x, x)$$

$$\Sigma \equiv \mathbb{R}^N$$

mathematical gamma: $a=1, \dots, n$

$$\gamma_a \equiv \rho \circ \gamma(e_a)$$

physical gamma: $\Gamma_a \equiv (-i)\gamma_a$

$$I_N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{N \times N} \text{ on } \mathbb{C}^N$$

$$\{\gamma_a, \gamma_a\} = -2\eta_{ab} I_N$$

$$\{\Gamma_a, \Gamma_a\} = 2\eta_{ab} I_N$$

Chirality operator in even dim:

* Suppose $n=s+t=2u$, A chirality element for $CL(s,t)$ is a Clifford element of the form:

$$\omega = \lambda \gamma(e_1) \dots \gamma(e_n) \in CL(s,t) \otimes \mathbb{C}$$

$\{e_1, \dots, e_n\}$: oriented, orthonormal basis in $\mathbb{R}^{s,t}$, $\lambda \in \mathbb{C}$

Claim: ω does not depend on $\{e_1, \dots, e_n\}$

proof: Suppose: $e'_i = A_{ij} e_j$, $A_{ij} \in SO(s,t)$

under a vector isomorphism

$$\psi: \Lambda^u \mathbb{R}^{s,t} \rightarrow CL(s,t)$$

$$\gamma(e_1) \dots \gamma(e_n) = \psi(e_1 \wedge \dots \wedge e_n)$$

$$\gamma(e'_1) \dots \gamma(e'_n) = \psi(e'_1 \wedge \dots \wedge e'_n)$$

$$e'_1 \wedge \dots \wedge e'_n = \det(A) e_1 \wedge \dots \wedge e_n$$

$$\therefore \gamma(e_1) \dots \gamma(e_n) = \gamma(e'_1) \dots \gamma(e'_n)$$

* every chirality element ω satisfies

$$\{\omega, \gamma(e_a)\} = 0$$

$$[\omega, \gamma(e_a) \cdot \gamma(e_b)] = 0 \quad \forall 1 \leq a, b \leq n$$

* If $\lambda^2 = (-1)^{n+1}$, then $\omega^2 = 1$, $\lambda = i^{k+t}$

Def: The chirality element (in even dim) $n=s+t=2u$ for $CL(s,t)$ is defined by

$$\omega = -i^{k+t} \gamma(e_1) \dots \gamma(e_n) \in CL(s,t) \otimes \mathbb{C}$$

* If $n+t$ is even $\Rightarrow \omega \in CL(s,t)$ (real)

* γ_a : mathematical gamma in the $\mathbb{C}$ rep of $CL(s,t)$:

$$\gamma_{n+1} \equiv -i^{k+t} \gamma_1 \dots \gamma_n \quad (\text{math. chirality op})$$

$$\Gamma_{n+1} \equiv -i^{k+t} \Gamma_1 \dots \Gamma_n \quad (\text{physical " "})$$

$$\gamma_{n+1} \equiv (-1)^k \Gamma_{n+1}?$$

* Character of Complex CL:

$CL(2k)$:

$$\omega = -i^k \gamma(e_1) \dots \gamma(e_{2k}) \in CL(2k)$$

Examples of C.A.: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\sigma_i^2 = I_2, \quad \sigma_i \sigma_{j+1} = i \sigma_{j+2} \pmod{3}$$

(i) $CL(1,0)$: $\{1, \gamma(e_1)\}$

$$\gamma(e_1)^2 = -1 \Rightarrow \gamma(e_1) \sim i$$

$$CL(1,0) \cong \mathbb{C}$$

(ii) $CL(0,1)$: $\gamma(e_1)^2 = 1$

• span $CL(0,1)$ with basis vectors $\frac{1}{2}(1 \pm \gamma(e_1))$

$\exists$ map $CL(0,1) \rightarrow \mathbb{R} \oplus \mathbb{R}$

$$\alpha \left[\frac{1}{2}(1 + \gamma(e_1)) \right] + \beta \left[\frac{1}{2}(1 - \gamma(e_1)) \right] \mapsto (\alpha, \beta)$$

$$\begin{aligned} \alpha' 1 + \beta' \gamma(e_1) &= \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \left(\frac{\alpha}{2} + \frac{\beta}{2} \right) 1 + \left(\frac{\alpha}{2} - \frac{\beta}{2} \right) \gamma(e_1) \end{aligned}$$

* Can find the respective basis by imposing

(a) $\beta' = 0, \quad \alpha' = 1$

$$\alpha = \beta \quad | \quad \alpha = 1$$

$$\therefore \hat{1} = (1, 1)$$

(b) $\alpha' = 0, \quad \beta' = 1$

$$\alpha = -\beta \Rightarrow \alpha' = 1 \Rightarrow \gamma(e_1) = (1, -1)$$

$$CL(0,1) \cong \mathbb{R} \oplus \mathbb{R} \quad \leftarrow \text{Basis } \frac{1}{2}(1 \pm e_1)$$

$$1 \mapsto (1, 1)$$

$$\gamma(e_1) \mapsto (1, -1)$$

(iii) $CL(1) \cong \mathbb{C} \otimes \mathbb{C} \cong CL(0,1) \otimes \mathbb{C}$

$$\text{Since } CL(0,1) \cong \mathbb{R} \oplus \mathbb{R} \Rightarrow (\mathbb{R} \oplus \mathbb{R}) \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C} \cong CL(1,0) \otimes \mathbb{C}$$

$$\frac{1}{2}(1 \pm i\gamma_1) \Rightarrow \hat{1} = (1, 1)$$

$$\gamma(e_1) = (i, -i)$$

(iv) $\gamma_1 \equiv i\sigma_1, \quad \gamma_2 \equiv i\sigma_2 \mid \gamma_1^2 = -I_2, \quad \gamma_2^2 = I_2, \quad \{\gamma_1, \gamma_2\} = 0$

generate a faithful rep of $CL(1,1)$ on $\mathbb{C}^2$

$$\dim(\mathbb{C}L(2,0)) = 2^2 = 4$$

$$\{1, \gamma_1, \gamma_2, \gamma_1 \gamma_2\}$$

$$(v) \{I_2, \sigma_1, \sigma_2, \sigma_3 \equiv -i\sigma_1\sigma_2\} \text{ span } \text{Mat}(2 \times 2, \mathbb{C}) \cong \text{End}(\mathbb{C}^2)$$

$$\mathbb{C}L(2) \cong \text{End}(\mathbb{C}^2)$$

$$(vi) \gamma_0 \equiv \begin{pmatrix} 0 & iI_2 \\ iI_2 & 0 \end{pmatrix}, \gamma_k \equiv \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \quad k=1,2,3$$

$$\mathbb{C}L(n) = \mathbb{C}L(n,0) \otimes_{\mathbb{R}} \mathbb{C}$$

$$(\gamma_0)^2 = -I_4$$

$$(\gamma_k)^2 = -I_4 \quad k=1,2,3$$

$$\gamma_a \gamma_b + \gamma_b \gamma_a = 0 \quad \forall a \neq b$$

$\Rightarrow$ Mathematical gamma matrices for $\mathbb{C}L(4,0)$

Another choice

$$\gamma_0 \equiv \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$$

$$\gamma_k = \begin{pmatrix} 0 & i\sigma_k \\ i\sigma_k & 0 \end{pmatrix} \quad k=1,2,3$$

$$k=1,2,3$$

$$\dim(\mathbb{C}L(4,0)) = 2^4 = 16$$

$$1$$

$$1$$

$$\gamma_a$$

$$4$$

$$\gamma_a \gamma_b$$

$$6$$

$$a < b$$

$$\gamma_a \gamma_b \gamma_c$$

$$4$$

$$a < b < c$$

$$\gamma_0 \gamma_1 \gamma_2 \gamma_3$$

$$1$$

* Physical Gamma of $\mathbb{C}L(1,3)$ in the Weyl/Dirac rep

$$\Gamma_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$$

$$\Gamma_k = \begin{pmatrix} 0 & \sigma_k \\ \sigma_k & 0 \end{pmatrix}, \quad \forall k=1,2,3$$

$$\bullet \Gamma'_a = \gamma_a = i\Gamma_a$$

$$\Gamma_0^2 = I_4$$

$$\gamma_a = i\Gamma'_a = -\Gamma_a$$

$$\Gamma_k^2 = -I_4$$

$\hookrightarrow$ Generate $(-+++)$

$$\bullet \mathbb{C}L(4) \cong \text{End}(\mathbb{C}^4)$$

* Complex Cliff. Algs are periodic:

$$\text{Lemma: } \mathbb{C}L(n+2) \cong \mathbb{C}L(n) \otimes \mathbb{C}L(2) \\ \cong \mathbb{C}L(n) \otimes \text{End}(\mathbb{C}^2)$$

$$\text{proof: write } \mathbb{C}^{n+2} \cong \mathbb{C}^n \otimes \mathbb{C}^2 = (x, y) \mid x \in \mathbb{C}^n, y \in \mathbb{C}^2$$

$$\bullet \text{Define: } S: \mathbb{C}^{n+2} \rightarrow \mathbb{C}L(n) \otimes \mathbb{C}L(2)$$

$$(x, y) \mapsto x \otimes \omega + 1 \otimes y$$

$$\bullet \text{where, for standard basis } \{e_1, e_2\} \text{ of } \mathbb{C}^2 \quad \omega = -i\gamma_1\gamma_2$$

$$S(x, y)^2 = (x \otimes \omega + 1 \otimes y) \cdot (x \otimes \omega + 1 \otimes y)$$

$$= x^2 \otimes \omega^2 + x \otimes \omega y + x \otimes y \omega + 1 \otimes y^2$$

$$\left\{ \begin{array}{l} \text{since } \{\omega, \alpha\} = 0 \quad \text{and} \quad \omega^2 = 1 \\ \Rightarrow x^2 \otimes 1 + x \otimes \omega y + x \otimes y \omega + 1 \otimes y^2 \end{array} \right.$$

$$= x^2 \otimes 1 + 1 \otimes y^2$$

$$x^2 = -g(x, x) \cdot 1, \quad y^2 = -\tilde{g}(y, y) \cdot 1$$

$$g(x, y)^2 = -g_{xx} \otimes 1 - 1 \otimes -g_{yy}$$

* Note from Complexification $(V, Q) \mapsto (V^{\mathbb{C}}, Q^{\mathbb{C}})$

where $V^{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$

then $Q^{\mathbb{C}}$ extends as follows

$$Q^{\mathbb{C}}: V^{\mathbb{C}} \times V^{\mathbb{C}} \rightarrow \mathbb{C}$$

$$Q^{\mathbb{C}}(V \otimes z, V \otimes z) := z^2 Q(V, V)$$

Thm: As complex algebras the $\mathbb{C}L$ and its even part are given:

n	$\mathbb{C}L(n)$	$\mathbb{C}L^0(n)$	N
Even	$\text{End}(\mathbb{C}^N)$	$\text{End}(\mathbb{C}^{N/2}) \oplus \text{End}(\mathbb{C}^{N/2})$	$2^{n/2}$
Odd	$\text{End}(\mathbb{C}^N) \oplus \text{End}(\mathbb{C}^N)$	$\text{End}(\mathbb{C}^N)$	$2^{(n-1)/2}$

• Since $\mathbb{C}L^0(n) \cong \mathbb{C}L(n-1)$

• $n=4$: $\mathbb{C}L(4) = \text{End}(\mathbb{C}^4)$, $\mathbb{C}L(3) = \mathbb{C}L^0(4)$

$$\mathbb{C}L^0(4) = \text{End}(\mathbb{C}^2) \oplus \text{End}(\mathbb{C}^2)$$

• $n=1$: $\mathbb{C}L(1) = \text{End}(\mathbb{C}) \oplus \text{End}(\mathbb{C}) = \mathbb{C} \oplus \mathbb{C}$

$$\mathbb{C}L^0(1) = \mathbb{C}$$

$$n=2: \mathbb{C}L(2) = \text{End}(\mathbb{C}^2), \mathbb{C}L^0(2) = \mathbb{C}L(1) = \mathbb{C} \oplus \mathbb{C}$$

$$n=3: (N=2) \mathbb{C}L(3) = \mathbb{C}L(4) = \text{End}(\mathbb{C}^2) \oplus \text{End}(\mathbb{C}^2)$$

$$\mathbb{C}L^0(3) = \text{End}(\mathbb{C}^2)$$

$$\leadsto \mathbb{C}L(4) = \text{End}(\mathbb{C}^4) = \mathbb{C}L(2) \otimes \mathbb{C}L(2) = \text{End}(\mathbb{C}^2) \otimes \text{End}(\mathbb{C}^2)$$

$$\leadsto \mathbb{C}L(3) = \text{End}(\mathbb{C}^2) \oplus \text{End}(\mathbb{C}^2) = \mathbb{C}L(1) \otimes \text{End}(\mathbb{C}^2) = [\mathbb{C} \oplus \mathbb{C}] \otimes \text{End}(\mathbb{C}^2)$$

thus, $\mathbb{C} \oplus \text{End}(\mathbb{C}^2)$ is trivially $\text{End}(\mathbb{C}^2)$

Thm: Structure thm of $\mathbb{R}$ Clifford Algebras: $\mathbb{C}L(s, t), \mathbb{C}L^0(s, t)$

$p \equiv s - t$, $n = s + t$ (all endomorphism algebras are understood as $\mathbb{R}$ -alg.)

$p \bmod(8)$	$\mathbb{C}L(s, t)$	N	$\mathbb{C}L^0(s, t)$	N
0	$\text{End}(\mathbb{R}^N)$	$2^{n/2}$	$\text{End}(\mathbb{R}^N) \oplus \text{End}(\mathbb{R}^N)$	$2^{(n-2)/2}$
1	$\text{End}(\mathbb{C}^N)$	$2^{(n-1)/2}$	$\text{End}(\mathbb{R}^N)$	$2^{(n-1)/2}$
2	$\text{End}(\mathbb{H}^N)$	$2^{(n-2)/2}$	$\text{End}(\mathbb{C}^N)$	$2^{(n-2)/2}$

3	$\text{End}(\mathbb{H}^N) \oplus \text{End}(\mathbb{H}^N)$	$2^{(n-3)/2}$	$\text{End}(\mathbb{H}^N)$	$2^{(n-3)/2}$
4	$\text{End}(\mathbb{H}^N)$	$2^{(n-2)/2}$	$\text{End}(\mathbb{H}^N) \oplus \text{End}(\mathbb{H}^N)$	$2^{(n-4)/2}$
5	$\text{End}(\mathbb{C}^N)$	$2^{(n-1)/2}$	$\text{End}(\mathbb{H}^N)$	$2^{(n-3)/2}$
6	$\text{End}(\mathbb{R}^N)$	$2^{n/2}$	$\text{End}(\mathbb{C}^N)$	$2^{(n-2)/2}$
7	$\text{End}(\mathbb{R}^N) \oplus \text{End}(\mathbb{R}^N)$	$2^{(n-1)/2}$	$\text{End}(\mathbb{R}^N)$	$2^{(n-1)/2}$

- $\text{CL}(1,3)$: $p = -2 \pmod{8} \rightarrow 6$ $n=4, N=4$
 $\text{CL}(1,3) \cong \text{End}(\mathbb{R}^4)$, $\text{CL}^0(1,3) \cong \text{End}(\mathbb{C}^2)$
 - $\text{CL}(3,1)$: $p = 2$, $n=4, N=2$
 $\text{CL}(3,1) \cong \text{End}(\mathbb{H}^2)$, $\text{CL}^0(3,1) \cong \text{End}(\mathbb{C}^2)$
- $\Rightarrow$ Hence, $\exists$ an isomorphism $\text{End}(\mathbb{H}^2) \longrightarrow \text{End}(\mathbb{R}^4)$

* Spinor Rep: (Dirac) spinor space $\Delta_n = \mathbb{C}^N$

$$\rho: \text{Cl}(n) \rightarrow \text{End}(\Delta_n)$$

• Def: The mathematical Clifford multiplication:

$$\mathbb{R}^{s,t} \times \Delta_n \longrightarrow \Delta_n$$

$$(X, \psi) \mapsto X \cdot \psi \equiv \rho(\gamma(X))\psi$$

physical "...":

$$(x, \psi) \mapsto x \cdot \psi \equiv \rho(\Gamma(x))\psi = -i \rho(\gamma(x))\psi$$

* From the isomorphism $\wedge^* \mathbb{R}^{s,t} \rightarrow \text{Cl}(s,t)$ allows us to define a (mathematical) Clift. mult. of spinors and forms:

$$\gamma(\mathbb{R}^{s,t}) \subset \text{Cl}(s,t), \quad e_a \in \mathbb{R}^{s,t}, \quad \psi \in \mathbb{C}^N$$

$$e_a \cdot \psi = \Gamma_a \psi \quad (\text{physical})$$

$$e_a \cdot \psi = \gamma_a \psi = i \Gamma_a \psi \quad (\text{mathematical})$$

• Recall $\text{Cl}^0(n) \cong \text{Cl}(n-1)$

* Induced Spinor reps on the Even Cl^0 :

• If n is odd, then induced reps is irreducible:

$$\text{Cl}^0(n) \xrightarrow{\cong} \text{End}(\Delta_n)$$

• If n is even, induced reps split into L & R handed Weyl spinors

$$\text{Cl}^0(n) \xrightarrow{\cong} \text{End}(\Delta_n^+) \oplus \text{End}(\Delta_n^-)$$

n	Induced reps	Spinor Space	N
Even	$\mathcal{CL}^0(n) \cong \text{End}(\Delta_n^+) \oplus \text{End}(\Delta_n^-)$	$\Delta_n^+ \cong \Delta_n^- \cong \mathbb{C}^{N/2}$	$2^{n/2}$
odd	$\mathcal{CL}^0(n) \cong \text{End}(\Delta_n)$	$\Delta_n \cong \mathbb{C}^N$	$2^{(n+1)/2}$

* Weyl Spinors & chirality in Even dim: $n=2d$, Δ_n the direct Rep and Γ_n the chirality element

(i) $\Delta_n^\pm$ can be identified w the (± 1) -eigen spaces of Γ_{n+1} on Δ_n

(ii) The induced reps of $\mathcal{CL}^0(n)$ maps $\Delta_n^\pm$ to itself, while elements in $\mathcal{CL}^1(n)$ (such as vectors in $\mathbb{R}^{d+1}$) map $\Delta_n^\pm$ to $\Delta_n^\mp$ and

$$\mathcal{CL}^0(n) \cong \text{Hom}(\Delta_n^+, \Delta_n^+) \oplus \text{Hom}(\Delta_n^-, \Delta_n^-)$$

$$\mathcal{CL}^1(n) \cong \text{Hom}(\Delta_n^+, \Delta_n^-) \oplus \text{Hom}(\Delta_n^-, \Delta_n^+)$$

proof: Note Γ_{n+1} is not idempotent since $\Gamma_{n+1}^2 = \mathbb{1}$

Let $\psi \in \Delta_n$, then

$$e_a \cdot \psi \rightarrow i \Gamma_a \psi$$

• $\Gamma_{n+1}^2 \psi = \psi$ and so ψ is an eigenvec of Γ_{n+1}^2 w e.v. 1

$$\Gamma_{n+1}^2 \psi = \lambda^2 \psi \Rightarrow \lambda = \pm 1$$

So Γ_{n+1}^2 splits $\Delta_n = \Delta_n^+ \oplus \Delta_n^-$

$\forall 1 \leq a, b \leq n$

• From above $[\Gamma_{n+1}, \Gamma_a \Gamma_b] = 0$ and $\{\Gamma_{n+1}, \Gamma_a\} = 0$

Spin Group: $\text{End}(\mathbb{K}^n) = \mathbb{K}^{n \times n}$

• Invertible: $GL(n, \mathbb{K}) \subset \text{End}(\mathbb{K}^n)$

Def: The group of invertible elements: in $CL(s, t)$

$$CL^\times(s, t) \equiv \{x \in CL(s, t) \mid \exists y \in CL(s, t) : xy = yx = 1\}$$

(same for $CL^\times(n)$)

Lemma: $CL^\times(s, t)$ is an open subset of $CL(s, t)$ and therefore a Lie group

Proof: $n = s + t$, $\mathcal{CL}(n) \cong \mathcal{CL}(s, t) \otimes_{\mathbb{R}} \mathbb{C}$

$$y \in \mathcal{CL}(n), \quad y = u + iv, \quad u, v \in \mathcal{CL}(s, t)$$

Follows that $CL^\times(s, t) = \mathcal{CL}^\times(n) \cap \mathcal{CL}(s, t)$

Where

$$\mathcal{CL}^\times(n) = \{x \in \mathcal{CL}(n) \mid \exists y \in \mathcal{CL}(n) : xy = yx = 1\}$$

→ Saying only the "real" part of $\mathcal{CL}^\times(n)$ is invertible?

$$\mathcal{CL}(n) \sim \text{End}(\mathbb{R}^N)$$

$$\text{Note } S_+^{s, t} \cong S^{n-1}$$

$$CL^x(n) \sim GL(n, \mathbb{C}) \quad \checkmark$$

Def: Subsets of $\mathbb{R}^{s,t}$:

$$\begin{aligned} S_+^{s,t} &\equiv \{v \in \mathbb{R}^{s,t} \mid \eta(v, v) = +1\} \\ S_-^{s,t} &\equiv \{v \in \mathbb{R}^{s,t} \mid \eta(v, v) = -1\} \\ S_{\pm}^{s,t} &\equiv S_+^{s,t} \cup S_-^{s,t} \end{aligned}$$

Prop: The following subsets of $CL(s, t)$ form subgroups of the group $CL^x(s, t)$

$$Pin(s, t) \equiv \{v_1, v_2, \dots, v_r \mid v_i \in S_{\pm}^{s,t}, r \geq 0\}$$

$$\begin{aligned} Spin(s, t) &\equiv Pin(s, t) \cap CL^0(s, t) \\ &= \{v_1, v_2, \dots, v_r \mid v_i \in S_{\pm}^{s,t}, r \geq 0\} \end{aligned}$$

$$Spin^+(s, t) \equiv \{v_1, \dots, v_{2p}, w_1, \dots, w_{2q} \mid v_i \in S_+^{s,t}, w_j \in S_-^{s,t}, p, q \geq 0\}$$

$$Pin(u) \equiv Pin(u, 0)$$

$$Spin(u) \equiv Spin(u, 0)$$

↑ (orthochronous)
(Spin group)

• Also:

$$Spin^+(s, t) = \{v_1, v_2, \dots, v_{2r} \mid r = p + q, 2q \text{ } v_i \in S_+^{s,t} \text{ and } 2p \text{ } v_i \in S_-^{s,t}\}$$

Let $v \in S_+^{s,t}$ and $w \in S_-^{s,t}$ and write $w = w_{||} + w_{\perp}$.

$$w_{||} = \eta(v, w)v, \quad \eta(v, w_{\perp}) = 0$$

$$\bullet \quad wv + vw = -2\eta(v, w)1$$

$$(w_{||} + w_{\perp})v + v(w_{||} + w_{\perp})$$

$$= \{w_{||}, v\} + \{w_{\perp}, v\}$$

$$= -2\eta(w_{||}, v)1$$

$$wv$$

Def: For $z \in v_1, v_2, \dots, v_r \in Pin(s, t)$

$$\deg(z) = \begin{cases} 0 & z \in CL^0(s, t) \\ 1 & z \in CL^1(s, t) \end{cases}$$

$$\deg^+(z) = \begin{cases} 0 & , \text{ even } \# v_i \in S_+^{s,t} \\ 1 & , \text{ odd } \# v_i \in S_+^{s,t} \end{cases}$$

$$\bullet \quad z \in Spin(s, t) \Rightarrow \deg(z) = 0$$

Let $\deg^+(z)$ deals with the transpose since

$$z^{-1} = v_r^{-1} \dots v_1^{-1} = (-1)^{\deg^+(z)} v_r \dots v_1$$

$$u = V_1 V_2 \Rightarrow u^{-1} = V_2^{-1} V_1^{-1}$$

$$\leadsto V_1 V_1^{-1} = 1 \Rightarrow V_1^{-1} = V_1 / -\mathcal{R}_1(V_1, V_1)$$

Def: Identify $\mathbb{R}^{s,t}$ in a canonical way via the embedding γ and

consider the map

$$R: \text{Pin}(s,t) \times \mathbb{R}^{s,t} \longrightarrow \mathbb{R}^{s,t}$$

$$(u, x) \mapsto (-1)^{\deg(u)} u x u^{-1}$$

The adjoint action

$\leadsto$ yields a continuous homo: $\lambda: \text{Pin}(s,t) \rightarrow \mathcal{O}(s,t)$

$$u \mapsto R_u \equiv R(u, \cdot)$$

$$* \text{ Ad}: V \times V \rightarrow V \quad u^{-1} = -u / \mathcal{Q}(u)$$

$$\text{for } u \in \text{Pin}(s,t) \quad \text{Ad}_u: V \rightarrow V, \quad x \in V$$

$$\text{Ad}_u(x) \equiv u x u^{-1} = -\frac{u x u}{\mathcal{Q}(u, u)}$$

For the embedding $V \hookrightarrow \text{Cl}(V) \quad x \in V \subset \text{Cl}(V)$

$$\begin{aligned} \text{Ad}_u(x) &= -\frac{1}{\mathcal{Q}_u} (u (-u x - 2\mathcal{Q}(x, u))) \\ &= \frac{1}{\mathcal{Q}(u, u)} [-\mathcal{Q}(u, u)x + 2\mathcal{Q}(x, u)u] \\ &= -x + 2 \frac{\mathcal{Q}(x, u)}{\mathcal{Q}(u, u)} u \\ &= -(x - 2 \frac{\mathcal{Q}(x, u)}{\mathcal{Q}(u, u)} u) \\ &= -R_u(x) \end{aligned}$$

R_u : Reflection on the hyperplane perpendicular to u , $x \in V \subset \text{Cl}(V)$
 $x - 2\mathcal{Q}(x, u) \hat{u}$

* Can extend this to Pin : $\text{Ad}_{u_1 \dots u_p} \equiv \text{Ad}_{u_1} \circ \dots \circ \text{Ad}_{u_p}$

$$\text{Ad}_{u_1 \dots u_p}: V \rightarrow V$$

$$\text{Twisted Adjoint: } \widetilde{\text{Ad}}_u(x) \equiv (-u) x u^{-1} = R_u(x)$$

$a \mapsto \tilde{a}$ (Grading automorphism of Cl)

$$a \mapsto (-1)^{\deg(a)} a$$

* Let $a = u_1 \dots u_r \in \text{Pin}(V)$, then

$$\widetilde{\text{Ad}}_a \equiv R_a \equiv R_{u_1} \circ \dots \circ R_{u_r}$$

$$\widetilde{\text{Ad}}_a \in \text{SO}(s,t)$$

• Note this is a group homomorphism $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$

$$\rho(a) \in \text{SO}(n)$$

$$\rho(a)(x) \equiv a x a^{-1}$$

Lemma: (1) R is well defined i.e. if two images in the set $\mathbb{R}^{s,t}$ of Cl

Lemma: If R is a reflection then $R(v) = v - 2(v \cdot n)n$ where n is the normal vector to the hyperplane

(2) $\forall v \in S^{n-1}$
 $R_v = R(v, \cdot) : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$
 is the reflection in the hyperplane $v^\perp \subset \mathbb{R}^{n+1}$

(3) R is a continuous homomorphism
 $\lambda : P_n(S, t) \rightarrow O(S, t)$

proof: $v \in \mathbb{R}^{n+1}, R(v, v) = \pm 1$, then $v^{-1} = \mp v$

$$R_v(x) = -v x v^{-1} = \pm v x v$$

$$R_v(x) = \begin{cases} -x & \text{if } x \parallel v \\ x & \text{if } x \perp v \end{cases}$$

$R_{v_1 \dots v_r}(x) = R_{v_1} \circ \dots \circ R_{v_r}$ and so $R(z, x) \subset \mathbb{R}^{n+1}$ p.h

* $v \in \mathbb{R}^{n+1}$ with $R(v, v) = \pm 1$ act as reflections on $\mathbb{R}^{n+1}$, and a general element of the pin group acts as a composition of reflections hence $\subset O(S, t)$

* Let us look at Real reps of the spin group

$$\text{Spin}(n) = \text{Spin}(n, 0), \quad \mathbb{R}^n$$

$$n=1: SO(1) = \{1\} \quad S^\pm = S^0$$

$$\begin{aligned} \text{Spin}(1) &= \{v_1 v_2 \dots v_r \mid v_i \in S^0, r \in \mathbb{Z} \setminus \{0\}\} \\ &= \{\epsilon_1 \epsilon_2 \dots \epsilon_r v_r \mid \epsilon_i = \pm 1, r \in \mathbb{Z} \setminus \{0\}\} \\ &= \{1, -1\} \end{aligned}$$

$$\boxed{\text{Spin}(1) \cong \mathbb{Z}_2}$$

$$n=2: SO(2) \cong U(1) \cong S^1, \quad S^\pm = S^1 \quad v_i \in S^1$$

* Frame Bundle:

$$P_x^{SO}(M) := \{h : \mathbb{R}^n \rightarrow T_x M \mid h \text{ orientation preserving isometry}\}$$

• Each $h \in P_x^{SO}(M)$ induces an orthonormal basis $h(e_1) \dots h(e_n)$ on $T_x M$

• $SO(n)$ acts on $P_x^{SO}(M)$

$$P_x^{SO}(M) \times SO(n) \rightarrow P_x^{SO}(M)$$

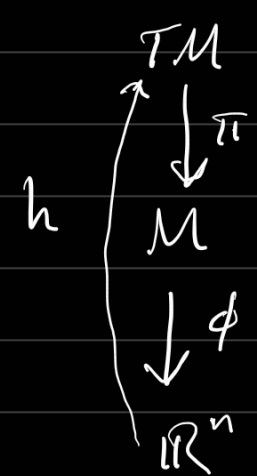
$$(h, A) \mapsto h \circ A$$

* This action is transitive: $\forall h_i, h_j \in P_x^{SO}$

$$\exists A \in SO(n) \Rightarrow h_i = h_j \circ A$$

$$A = h_i^{-1} \circ h_j$$

* For fixed (frame) $h_0 \in P_x^{SO}(M)$



$$SO(n) \rightarrow P_x^{so}(M)$$

$$A \mapsto h \circ A$$

is bijective.

$$P^{so}(M) := \bigsqcup_{x \in M} P_x^{so}(M)$$

(Frame Bundle)

* Note, If $v \in \mathbb{R}^n$, then $v = v^i e_i$, let x^i define a basis on $T_x M$. Then

$$h(v) = h(v^i e_i) = v^i h(e_i)$$

$$h(e_i) := X_i$$

$$h(v) = v^i X_i$$

* Given a frame $(v_1, \dots, v_n) \in P^{so}(M)$, then

$$v \circ A = (v_i A_{i1}, v_i A_{i2}, \dots, v_i A_{in})$$

* Note this is just matrix mult.

$$\tilde{z}^i = A^i_j v^j \Rightarrow \tilde{z}^T = A^T_j v^j$$

Defining a linear map as an element of a $(1,1)$ tensor

$$\omega : V^* \times V \longrightarrow V$$

$$\left\{ \begin{array}{l} \omega(a, \cdot) \equiv L : V \rightarrow V \end{array} \right.$$

$$\left\{ \begin{array}{l} L(v) = \frac{v^m L(a_m)}{v^T L v} = \frac{v^m X_m}{v^T L v} \Rightarrow v^T L \equiv \omega(\tilde{v}^T, \cdot) \end{array} \right.$$

oops, should be in Bundle notes